

Embedded Software Engineer

Ju Ta Tao (Harry)

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Embedded software engineer focused on RTOS, OT protocols, and system architecture.



Work Experience

Moxa Inc. | Software Engineer R&D

2023 - Present

- Designed embedded OT and protocol architectures.
- Developed P2P and Modbus communication frameworks.
- Maintained CI/CD workflows for legacy products.
- Optimized system reliability on constrained platforms.
- Resolved customer issues across protocol and embedded systems.
- Worked with PM, QA, and front-end teams on embedded products.

Academia Sinica | Intern

Jul. 2019 - Sep. 2019

- Developed speech recognition web applications.

Skills

C / C++ / Python

Embedded Linux

Zephyr

FreeRTOS

Modbus TCP

MQTT

SNMP

CI/CD

Docker

ARM

AI / Deep Learning

Education

National Chung Cheng University

2020 - 2022

M.S. in Computer Science and Information Engineering

- AI and speech recognition research using PyTorch.
- Focused on low-resource speech recognition systems.

Chang Gung University

2015 - 2020

B.S. in Computer Science and Information Engineering

- Built AI and speech recognition projects using TensorFlow and Django.
- Developed embedded and RTOS-based ARM projects.

Projects

Decentralized MQTT Broker Mesh

github.com/judadao/mqtt_field_bridge_app

High-availability MQTT broker mesh for Zephyr, ESP32, and Linux edge deployments.

Core Features

- Lightweight MQTT broker running on Zephyr RTOS, ESP32, and Linux.
- Broker Mesh that synchronizes publish/subscribe traffic across nodes.
- Failover and fallback path that keeps clients online during broker failures.
- Automatic load balancing for client connections and topic subscriptions.

Results

- Validated with ESP32 Wi-Fi brokers, ESP32 + W5500 Ethernet brokers, and Linux brokers.
- Reached 99.99% delivery in fixed-message broker failure recovery tests.
- Matched Mosquitto-level throughput in single-broker benchmark testing.